

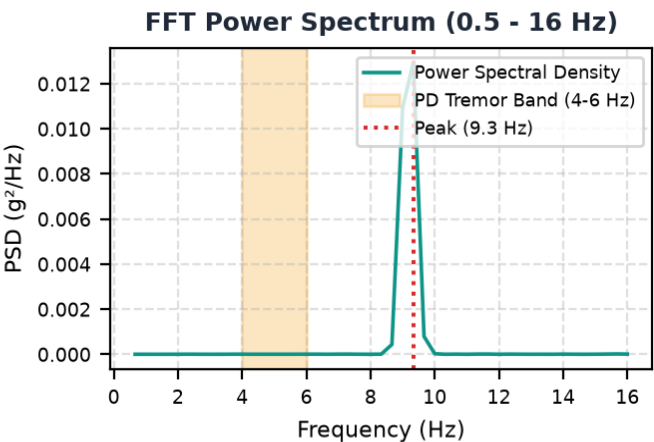
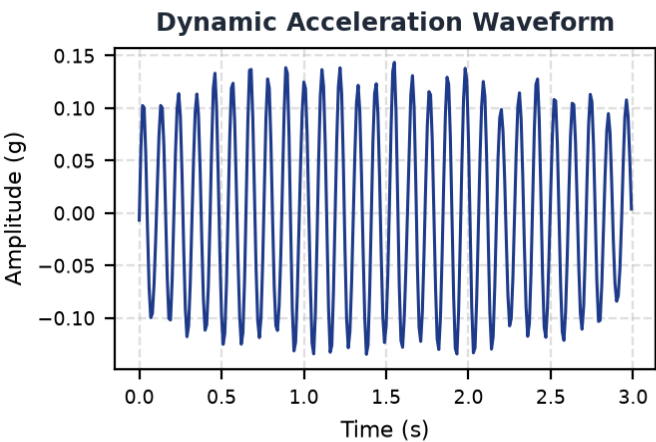
NEUROTRACK — Tremor Assessment & Telemetry Report

Report ID: NT-1788551963
Date: 2026-09-05 01:29:23

Patient Identifier:	ET_01	Data Origin:	Synthetic Fallback Dataset
Sensor Device:	MPU6050 6-DoF IMU Ring	Sampling Rate:	100.0 Hz (Nyquist Band: 50 Hz)
Assessment Duration:	60.0 seconds	Analysis Window:	3.0 sec (50% sliding overlap)

AI Pattern Classification	Tremor Severity Index	Clinical Severity Grade
PD Confidence: 57.6% (PD Match: 57.6%)	29.2 / 100	Mild Low-amplitude periodic oscillation detected in 4-6 Hz band with intermittent damping.

Biomechanical Time-Series & Spectral Isolation



Extracted Digital Biomarkers (Window Metrics)

Biomarker	Observed Value	Reference Band / Diagnostic Context
Dominant Frequency	9.33 Hz	Parkinsonian Resting Tremor: 4.0 - 6.0 Hz
Tremor Band Power Ratio	0.1%	Ratio of power in 4-6 Hz band to broad spectrum (>40% indicates localized tremor)
Dynamic Jerk RMS	4.97 g/s	Time-derivative of acceleration measuring abrupt oscillatory changes
Spectral Shannon Entropy	0.247	Low (<0.45) indicates rhythmic periodicity; High (>0.70) indicates random/voluntary noise

Physiological Interpretation & Explainability

Dominant frequency (9.3 Hz) is elevated above the Parkinsonian resting band, more characteristic of postural/action tremors (e.g., Essential Tremor) or physiological tremor. Tremor-band energy is minimal (0.1%), consistent with physiological resting state. Low spectral entropy (0.25) indicates high wave regularity, reflecting rhythmic involuntary motor firing.

NOTICE & MEDICAL DISCLAIMER: *NeuroTrack is an experimental screening and longitudinal monitoring aid, NOT a diagnostic or medical treatment device. The metrics, classifications, and severity scores provided herein reflect biomechanical time-series observations and must be interpreted solely by a qualified neurologist or physician in clinical context. Never initiate, alter, or discontinue any medication based on this report.*